

Aufgabe 1

1.1

Befehl	Ausgabe																																																																																																									
<pre>SELECT employee_id AS EmpID, last_name AS Employee, job_id AS Job, hire_date AS HireDate FROM employees;</pre>	<table><tr><th></th><th>EMPID</th><th>EMPLOYEE</th><th>JOB</th><th>HIREDATE</th></tr><tr><td>1</td><td>100</td><td>King</td><td>AD_PRES</td><td>17-JUN-87</td></tr><tr><td>2</td><td>101</td><td>Kochhar</td><td>AD_VP</td><td>21-SEP-89</td></tr><tr><td>3</td><td>102</td><td>De Haan</td><td>AD_VP</td><td>13-JAN-93</td></tr><tr><td>4</td><td>103</td><td>Hunold</td><td>IT_PROG</td><td>03-JAN-90</td></tr><tr><td>5</td><td>104</td><td>Ernst</td><td>IT_PROG</td><td>21-MAY-91</td></tr><tr><td>6</td><td>107</td><td>Lorentz</td><td>IT_PROG</td><td>07-FEB-99</td></tr><tr><td>7</td><td>124</td><td>Mourgos</td><td>ST_MAN</td><td>16-NOV-99</td></tr><tr><td>8</td><td>141</td><td>Rajs</td><td>ST_CLERK</td><td>17-OCT-95</td></tr><tr><td>9</td><td>142</td><td>Davies</td><td>ST_CLERK</td><td>29-JAN-97</td></tr><tr><td>10</td><td>143</td><td>Matos</td><td>ST_CLERK</td><td>15-MAR-98</td></tr><tr><td>11</td><td>144</td><td>Vargas</td><td>ST_CLERK</td><td>09-JUL-98</td></tr><tr><td>12</td><td>149</td><td>Zlotkey</td><td>SA_MAN</td><td>29-JAN-00</td></tr><tr><td>13</td><td>174</td><td>Abel</td><td>SA_REP</td><td>11-MAY-96</td></tr><tr><td>14</td><td>176</td><td>Taylor</td><td>SA_REP</td><td>24-MAR-98</td></tr><tr><td>15</td><td>178</td><td>Grant</td><td>SA_REP</td><td>24-MAY-99</td></tr><tr><td>16</td><td>200</td><td>Whalen</td><td>AD_ASST</td><td>17-SEP-87</td></tr><tr><td>17</td><td>201</td><td>Hartstein</td><td>MK_MAN</td><td>17-FEB-96</td></tr><tr><td>18</td><td>202</td><td>Fay</td><td>MK_REP</td><td>17-AUG-97</td></tr><tr><td>19</td><td>205</td><td>Higgins</td><td>AC_MGR</td><td>07-JUN-94</td></tr><tr><td>20</td><td>206</td><td>Gietz</td><td>AC_ACCOUNT</td><td>07-JUN-94</td></tr></table>		EMPID	EMPLOYEE	JOB	HIREDATE	1	100	King	AD_PRES	17-JUN-87	2	101	Kochhar	AD_VP	21-SEP-89	3	102	De Haan	AD_VP	13-JAN-93	4	103	Hunold	IT_PROG	03-JAN-90	5	104	Ernst	IT_PROG	21-MAY-91	6	107	Lorentz	IT_PROG	07-FEB-99	7	124	Mourgos	ST_MAN	16-NOV-99	8	141	Rajs	ST_CLERK	17-OCT-95	9	142	Davies	ST_CLERK	29-JAN-97	10	143	Matos	ST_CLERK	15-MAR-98	11	144	Vargas	ST_CLERK	09-JUL-98	12	149	Zlotkey	SA_MAN	29-JAN-00	13	174	Abel	SA_REP	11-MAY-96	14	176	Taylor	SA_REP	24-MAR-98	15	178	Grant	SA_REP	24-MAY-99	16	200	Whalen	AD_ASST	17-SEP-87	17	201	Hartstein	MK_MAN	17-FEB-96	18	202	Fay	MK_REP	17-AUG-97	19	205	Higgins	AC_MGR	07-JUN-94	20	206	Gietz	AC_ACCOUNT	07-JUN-94
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1.2

Befehl	Ausgabe																		
<pre>SELECT DISTINCT manager_id FROM employees WHERE manager_id IS NOT NULL;</pre>	<table> <tr> <th></th><th>MANAGER_ID</th></tr> <tr><td>1</td><td>124</td></tr> <tr><td>2</td><td>103</td></tr> <tr><td>3</td><td>101</td></tr> <tr><td>4</td><td>149</td></tr> <tr><td>5</td><td>201</td></tr> <tr><td>6</td><td>100</td></tr> <tr><td>7</td><td>102</td></tr> <tr><td>8</td><td>205</td></tr> </table>		MANAGER_ID	1	124	2	103	3	101	4	149	5	201	6	100	7	102	8	205
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3	101																		
4	149																		
5	201																		
6	100																		
7	102																		
8	205																		

1.3

Befehl	Ausgabe										
<pre>SELECT * FROM departments WHERE manager_id = 201;</pre>	<table><tr><th></th><th>DEPARTMENT_ID</th><th>DEPARTMENT_NAME</th><th>MANAGER_ID</th><th>LOCATION_ID</th></tr><tr><td>1</td><td>20</td><td>Marketing</td><td>201</td><td>1800</td></tr></table>		DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID	1	20	Marketing	201	1800
	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID							
1	20	Marketing	201	1800							

1.4

Befehl	Ausgabe																																																
<pre>SELECT last_name, salary FROM employees WHERE salary > 6000;</pre>	<table><tr><th></th><th>LAST_NAME</th><th>SALARY</th><th></th></tr><tr><td>1</td><td>King</td><td>24000</td><td></td></tr><tr><td>2</td><td>Kochhar</td><td>17000</td><td></td></tr><tr><td>3</td><td>De Haan</td><td>17000</td><td></td></tr><tr><td>4</td><td>Hunold</td><td>9000</td><td></td></tr><tr><td>5</td><td>Zlotkey</td><td>10500</td><td></td></tr><tr><td>6</td><td>Abel</td><td>11000</td><td></td></tr><tr><td>7</td><td>Taylor</td><td>8600</td><td></td></tr><tr><td>8</td><td>Grant</td><td>7000</td><td></td></tr><tr><td>9</td><td>Hartstein</td><td>13000</td><td></td></tr><tr><td>10</td><td>Higgins</td><td>12000</td><td></td></tr><tr><td>11</td><td>Gietz</td><td>8300</td><td></td></tr></table>		LAST_NAME	SALARY		1	King	24000		2	Kochhar	17000		3	De Haan	17000		4	Hunold	9000		5	Zlotkey	10500		6	Abel	11000		7	Taylor	8600		8	Grant	7000		9	Hartstein	13000		10	Higgins	12000		11	Gietz	8300	
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11	Gietz	8300																																															

1.5

Befehl	Ausgabe			
<pre>SELECT last_name, salary, department_id FROM employees WHERE employee_id = 124;</pre>		LAST_NAME	SALARY	DEPARTMENT_ID
	1	Mourgos	5800	50

1.6

Befehl	Ausgabe																																																																																				
<pre>SELECT last_name, job_id, hire_date FROM employees ORDER BY hire_date ASC;</pre>	<table><tr><th></th><th>LAST_NAME</th><th>JOB_ID</th><th>HIRE_DATE</th></tr><tr><td>1</td><td>King</td><td>AD_PRES</td><td>17-JUN-87</td></tr><tr><td>2</td><td>Whalen</td><td>AD_ASST</td><td>17-SEP-87</td></tr><tr><td>3</td><td>Kochhar</td><td>AD_VP</td><td>21-SEP-89</td></tr><tr><td>4</td><td>Hunold</td><td>IT_PROG</td><td>03-JAN-90</td></tr><tr><td>5</td><td>Ernst</td><td>IT_PROG</td><td>21-MAY-91</td></tr><tr><td>6</td><td>De Haan</td><td>AD_VP</td><td>13-JAN-93</td></tr><tr><td>7</td><td>Higgins</td><td>AC_MGR</td><td>07-JUN-94</td></tr><tr><td>8</td><td>Gietz</td><td>AC_ACCOUNT</td><td>07-JUN-94</td></tr><tr><td>9</td><td>Rajs</td><td>ST_CLERK</td><td>17-OCT-95</td></tr><tr><td>10</td><td>Hartstein</td><td>MK_MAN</td><td>17-FEB-96</td></tr><tr><td>11</td><td>Abel</td><td>SA_REP</td><td>11-MAY-96</td></tr><tr><td>12</td><td>Davies</td><td>ST_CLERK</td><td>29-JAN-97</td></tr><tr><td>13</td><td>Fay</td><td>MK_REP</td><td>17-AUG-97</td></tr><tr><td>14</td><td>Matos</td><td>ST_CLERK</td><td>15-MAR-98</td></tr><tr><td>15</td><td>Taylor</td><td>SA_REP</td><td>24-MAR-98</td></tr><tr><td>16</td><td>Vargas</td><td>ST_CLERK</td><td>09-JUL-98</td></tr><tr><td>17</td><td>Lorentz</td><td>IT_PROG</td><td>07-FEB-99</td></tr><tr><td>18</td><td>Grant</td><td>SA_REP</td><td>24-MAY-99</td></tr><tr><td>19</td><td>Mourgos</td><td>ST_MAN</td><td>16-NOV-99</td></tr><tr><td>20</td><td>Zlotkey</td><td>SA_MAN</td><td>29-JAN-00</td></tr></table>		LAST_NAME	JOB_ID	HIRE_DATE	1	King	AD_PRES	17-JUN-87	2	Whalen	AD_ASST	17-SEP-87	3	Kochhar	AD_VP	21-SEP-89	4	Hunold	IT_PROG	03-JAN-90	5	Ernst	IT_PROG	21-MAY-91	6	De Haan	AD_VP	13-JAN-93	7	Higgins	AC_MGR	07-JUN-94	8	Gietz	AC_ACCOUNT	07-JUN-94	9	Rajs	ST_CLERK	17-OCT-95	10	Hartstein	MK_MAN	17-FEB-96	11	Abel	SA_REP	11-MAY-96	12	Davies	ST_CLERK	29-JAN-97	13	Fay	MK_REP	17-AUG-97	14	Matos	ST_CLERK	15-MAR-98	15	Taylor	SA_REP	24-MAR-98	16	Vargas	ST_CLERK	09-JUL-98	17	Lorentz	IT_PROG	07-FEB-99	18	Grant	SA_REP	24-MAY-99	19	Mourgos	ST_MAN	16-NOV-99	20	Zlotkey	SA_MAN	29-JAN-00
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20	Zlotkey	SA_MAN	29-JAN-00																																																																																		

1.7

1.7

Befehl	Ausgabe									
<pre>SELECT last_name, department_id FROM employees WHERE department_id = 20 ORDER BY last_name ASC;</pre>	<table><tr><th></th><th>LAST_NAME</th><th>DEPARTMENT_ID</th></tr><tr><td>1</td><td>Fay</td><td>20</td></tr><tr><td>2</td><td>Hartstein</td><td>20</td></tr></table>		LAST_NAME	DEPARTMENT_ID	1	Fay	20	2	Hartstein	20
	LAST_NAME	DEPARTMENT_ID								
1	Fay	20								
2	Hartstein	20								

1.8

Befehl	Ausgabe																				
<pre>SELECT last_name AS Employee, salary AS MonthlySalary, commission_pct AS Commission FROM employees WHERE commission_pct = 0.2 OR job_id = 'SA_REP';</pre>	<table><tr><th></th><th>EMPLOYEE</th><th>MONTHLYSALARY</th><th>COMMISSION</th></tr><tr><td>1</td><td>Zlotkey</td><td>10500</td><td>0.2</td></tr><tr><td>2</td><td>Abel</td><td>11000</td><td>0.3</td></tr><tr><td>3</td><td>Taylor</td><td>8600</td><td>0.2</td></tr><tr><td>4</td><td>Grant</td><td>7000</td><td>0.15</td></tr></table>		EMPLOYEE	MONTHLYSALARY	COMMISSION	1	Zlotkey	10500	0.2	2	Abel	11000	0.3	3	Taylor	8600	0.2	4	Grant	7000	0.15
	EMPLOYEE	MONTHLYSALARY	COMMISSION																		
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4	Grant	7000	0.15																		

1.9

Befehl	Ausgabe															
<pre>SELECT first_name AS name, last_name AS surname, employee_id, salary -- restliche Namen sind passend FROM employees WHERE department_id = 60 AND manager_id = 103;</pre>	<table><tr><th></th><th>NAME</th><th>SURNAME</th><th>EMPLOYEE_ID</th><th>SALARY</th></tr><tr><td>1</td><td>Bruce</td><td>Ernst</td><td>104</td><td>6000</td></tr><tr><td>2</td><td>Diana</td><td>Lorentz</td><td>107</td><td>4200</td></tr></table>		NAME	SURNAME	EMPLOYEE_ID	SALARY	1	Bruce	Ernst	104	6000	2	Diana	Lorentz	107	4200
	NAME	SURNAME	EMPLOYEE_ID	SALARY												
1	Bruce	Ernst	104	6000												
2	Diana	Lorentz	107	4200												

Aufgabe 2

2.1

Welche Probleme können bei der sorglosen Verwendung von NATURAL JOIN entstehen?

NATURAL JOIN fügt auf Basis von gleichnamigen Attributen zusammen. Wenn mehrere gleichnamige Attribute vorhanden sind, werden immer alle geprüft. Evtl. ist das zum Erstellungszeitpunkt der Datenbank noch kein Problem, aber wenn später in eine der Tabellen ein Attribut mit einem gleichen Namen von der ersten Tabelle eingefügt wird oder ein Attribut umbenannt wird, kommt es sehr wahrscheinlich zu unvorhersehbarem Verhalten.

Befehl					
<pre>SELECT l.location_id AS location, l.street_address AS street, l.city AS city, l.state_province AS province, c.country_name AS country FROM locations l NATURAL JOIN countries c;</pre>					
Ausgabe					
	LOCATION	STREET	CITY	PROVINCE	COUNTRY
1	1400	2014 Jabberwocky Rd	Southlake	Texas	United States of America
2	1500	2011 Interiors Blvd	South San Francisco	California	United States of America
3	1700	2004 Charade Rd	Seattle	Washington	United States of America
4	1800	147 Spadina Ave	Toronto	Ontario	Canada
5	2500	Magdalen Centre, The Oxford Science Park	Oxford	Oxford	United Kingdom

2.2

Befehl	Ausgabe																																																																																
<pre>SELECT last_name, department_id, department_name FROM employees JOIN departments USING (department_id) ORDER BY last_name DESC;</pre>	<table><tr><th></th><th>LAST_NAME</th><th>DEPARTMENT_ID</th><th>DEPARTMENT_NAME</th></tr><tr><td>1</td><td>Zlotkey</td><td>80</td><td>Sales</td></tr><tr><td>2</td><td>Whalen</td><td>10</td><td>Administration</td></tr><tr><td>3</td><td>Vargas</td><td>50</td><td>Shipping</td></tr><tr><td>4</td><td>Taylor</td><td>80</td><td>Sales</td></tr><tr><td>5</td><td>Rajs</td><td>50</td><td>Shipping</td></tr><tr><td>6</td><td>Mourgos</td><td>50</td><td>Shipping</td></tr><tr><td>7</td><td>Matos</td><td>50</td><td>Shipping</td></tr><tr><td>8</td><td>Lorentz</td><td>60</td><td>IT</td></tr><tr><td>9</td><td>Kochhar</td><td>90</td><td>Executive</td></tr><tr><td>10</td><td>King</td><td>90</td><td>Executive</td></tr><tr><td>11</td><td>Hunold</td><td>60</td><td>IT</td></tr><tr><td>12</td><td>Higgins</td><td>110</td><td>Accounting</td></tr><tr><td>13</td><td>Hartstein</td><td>20</td><td>Marketing</td></tr><tr><td>14</td><td>Gietz</td><td>110</td><td>Accounting</td></tr><tr><td>15</td><td>Fay</td><td>20</td><td>Marketing</td></tr><tr><td>16</td><td>Ernst</td><td>60</td><td>IT</td></tr><tr><td>17</td><td>De Haan</td><td>90</td><td>Executive</td></tr><tr><td>18</td><td>Davies</td><td>50</td><td>Shipping</td></tr><tr><td>19</td><td>Abel</td><td>80</td><td>Sales</td></tr></table>		LAST_NAME	DEPARTMENT_ID	DEPARTMENT_NAME	1	Zlotkey	80	Sales	2	Whalen	10	Administration	3	Vargas	50	Shipping	4	Taylor	80	Sales	5	Rajs	50	Shipping	6	Mourgos	50	Shipping	7	Matos	50	Shipping	8	Lorentz	60	IT	9	Kochhar	90	Executive	10	King	90	Executive	11	Hunold	60	IT	12	Higgins	110	Accounting	13	Hartstein	20	Marketing	14	Gietz	110	Accounting	15	Fay	20	Marketing	16	Ernst	60	IT	17	De Haan	90	Executive	18	Davies	50	Shipping	19	Abel	80	Sales
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2.3

Befehl

```
SELECT last_name, job_id, department_id, department_name
FROM (employees JOIN departments USING (department_id))
JOIN locations USING (location_id)
WHERE UPPER(city) = 'OXFORD';
```

Ausgabe

	LAST_NAME	JOB_ID	DEPARTMENT_ID	DEPARTMENT_NAME
1	Zlotkey	SA_MAN	80	Sales
2	Abel	SA_REP	80	Sales
3	Taylor	SA_REP	80	Sales

2.4

Befehl

```
SELECT e.last_name AS Employee,
e.employee_id AS EmpId,
m.last_name AS Manager,
m.employee_id AS MgrId
FROM employees e JOIN employees m ON e.manager_id = m.employee_id
WHERE e.manager_id IS NOT NULL;
```

Ausgabe

	EMPLOYEE	EMPID	MANAGER	MGRID
1	Kochhar	101	King	100
2	De Haan	102	King	100
3	Hartstein	201	King	100
4	Zlotkey	149	King	100
5	Mourgos	124	King	100
6	Higgins	205	Kochhar	101
7	Whalen	200	Kochhar	101
8	Hunold	103	De Haan	102
9	Ernst	104	Hunold	103
10	Lorentz	107	Hunold	103
11	Rajs	141	Mourgos	124
12	Vargas	144	Mourgos	124
13	Davies	142	Mourgos	124
14	Matos	143	Mourgos	124
15	Grant	178	Zlotkey	149
16	Abel	174	Zlotkey	149
17	Taylor	176	Zlotkey	149
18	Fay	202	Hartstein	201
19	Gietz	206	Higgins	205

2.5

Befehl

```
SELECT e.last_name, e.salary
FROM employees e JOIN job_grades g ON e.salary BETWEEN g.lowest_sal AND g.highest_sal
WHERE grade_level = 'E';
```

Ausgabe

	LAST_NAME	SALARY
1	King	24000
2	Kochhar	17000
3	De Haan	17000

2.6

Befehl

```
SELECT e.last_name, e.job_id, d.department_name
FROM employees e LEFT OUTER JOIN departments d USING (department_id)
WHERE salary > 5000;
```

Ausgabe

	LAST_NAME	JOB_ID	DEPARTMENT_NAME
1	Hartstein	MK_MAN	Marketing
2	Fay	MK_REP	Marketing
3	Mourgos	ST_MAN	Shipping
4	Hunold	IT_PROG	IT
5	Ernst	IT_PROG	IT
6	Zlotkey	SA_MAN	Sales
7	Abel	SA_REP	Sales
8	Taylor	SA_REP	Sales
9	King	AD_PRES	Executive
10	Kochhar	AD_VP	Executive
11	De Haan	AD_VP	Executive
12	Higgins	AC_MGR	Accounting
13	Gietz	AC_ACCOUNT	Accounting
14	Grant	SA_REP	(null)

Aufgabe 3

3.1

Befehl

```
SELECT employee_id
FROM employees
MINUS
SELECT manager_id
FROM employees;
```

Ausgabe

	EMPLOYEE_ID
1	104
2	107
3	141
4	142
5	143
6	144
7	174
8	176
9	178
10	200
11	202
12	206

3.2

Befehl		
<pre>SELECT country_id, country_name FROM countries MINUS SELECT country_id, country_name FROM departments JOIN locations USING (location_id) JOIN countries USING (country_id);</pre>		
Ausgabe		
	COUNTRY_ID	COUNTRY_NAME
1	DE	Germany

3.3

Befehl	
<pre>SELECT employee_id FROM job_history UNION SELECT manager_id FROM employees WHERE manager_id IS NOT NULL;</pre>	
Ausgabe	
	EMPLOYEE_ID
1	100
2	101
3	102
4	103
5	124
6	149
7	176
8	200
9	201
10	205

3.4

Befehl		
<pre>SELECT employee_id, job_id FROM job_history INTERSECT SELECT employee_id, job_id FROM employees;</pre>		
Ausgabe		
	EMPLOYEE_ID	JOB_ID
1	176	SA_REP
2	200	AD_ASST

3.5

Befehl

```
SELECT employee_id  
FROM employees  
MINUS  
SELECT employee_id  
FROM employees JOIN departments USING (department_id)  
JOIN locations USING (location_id)  
WHERE city IN ('Seattle', 'Southlake');
```

Ausgabe

	EMPLOYEE_ID
1	124
2	141
3	142
4	143
5	144
6	149
7	174
8	176
9	178
10	201
11	202

Aufgabe 4

4.1

Befehl		
<pre>SELECT VorlNr, Titel FROM Studenten JOIN hoeren USING (MatrNr) JOIN Vorlesungen USING (VorlNr) WHERE Name = 'Carnap';</pre>		
Ausgabe		
	VORLNR	TITEL
1	5041	Ethik
2	5052	Wissenschaftstheorie
3	5216	Bioethik
4	5259	Der Wiener Kreis

4.2

Befehl	
<pre>SELECT v2.Titel FROM Vorlesungen v1 -- v1 ist der Nachfolger JOIN voraussetzen vr ON v1.VorlNr = vr.Nachfolger JOIN Vorlesungen v2 ON vr.Vorgaenger = v2.VorlNr -- v2 ist der Vorgänger WHERE v1.Titel = 'Ethik';</pre>	
Ausgabe	
	TITEL
1	Grundzuege

4.3

Befehl	
<pre>SELECT PersNr FROM Professoren MINUS SELECT Boss FROM Assistenten;</pre>	
Ausgabe	
	PERSNR
1	2133
2	2136
3	2137